

End of Encryption/Decryption (RSA Algorithm)

Estimated duration: 5 minutes

Introduction

This lab will guide you through the installation and initial use of a cryptographic tool called CrypTool. It focuses on encryption, decryption, and key generation using the RSA algorithm.

Learning objectives

- Install **CrypTool 2.1 (Stable Build 9778.2)** on your system
- Generate RSA keys and encrypt/decrypt a message

Part 1: Installing CrypTool 2.1

Step 1: Download CrypTool 2.1

- **CrypTool 2.1** is the current version available as a desktop application.
- Go to the official CrypTool website: <https://www.cryptool.org/en/>.
(*Note: To open the links, right-click (or long-press) on the links and select "Open in new tab." Avoid clicking the link directly, as this might block it.*)

Step 2: Install CrypTool 2.1 on Windows

- Download **CrypTool 2.1 (Stable Build 9778.2)** from the official website:
 - Go to [CrypTool 2 Download Page](#).
(*Note: To open the links, right-click (or long-press) on the links and select "Open in new tab." Avoid clicking the link directly, as this might block it.*)
 - Select the version compatible with your operating system (typically a .exe file for Windows).
- Run the installer by double-clicking the downloaded file.
 - Follow the prompts and accept the default installation options.

Part 2: Using RSA algorithm for encryption/decryption

Step 1: Generating an RSA key pair

- Open **CrypTool 2.1**.

- Select the **New** option.

CrypTool 2.1 (Stable Build 9778.2) – Startcenter

Home Edit Crypto Tutorials

 New ▾

 Open ▾

 Save ▾

 Print

 Play

 Stop

 Log

File Execute View

Startcenter Main control for CrypTool2 workspaces

★ Main functions

 Create a new workspace with the graphical editor

 Use the wizard to easily try some CrypTool 2 features

External resources

 View a tutorial video about CrypTool 2 on YouTube

 Visit the CrypTool Book page on the CrypTool portal

 Visit the official CrypTool 2 website

 Visit the official YouTube channel: You will learn how to use CrypTool 2 and how to add your own functions

 Visit us on Facebook

Templates (double click to open)

277

 Cryptography

 Cryptanalysis

 Hash Functions

 Mathematics

 Codes

 Protocols

 Steganography

 Tools

Recently opened workspaces (double click to open)

 RSA

 RSA Decryption

 RSA

 RSA Encryption

 Key

 RSA Key Generator

 AES

 AES (CBC) with PKCS#5 and IV

 DES

 DES Known-Plaintext Analysis

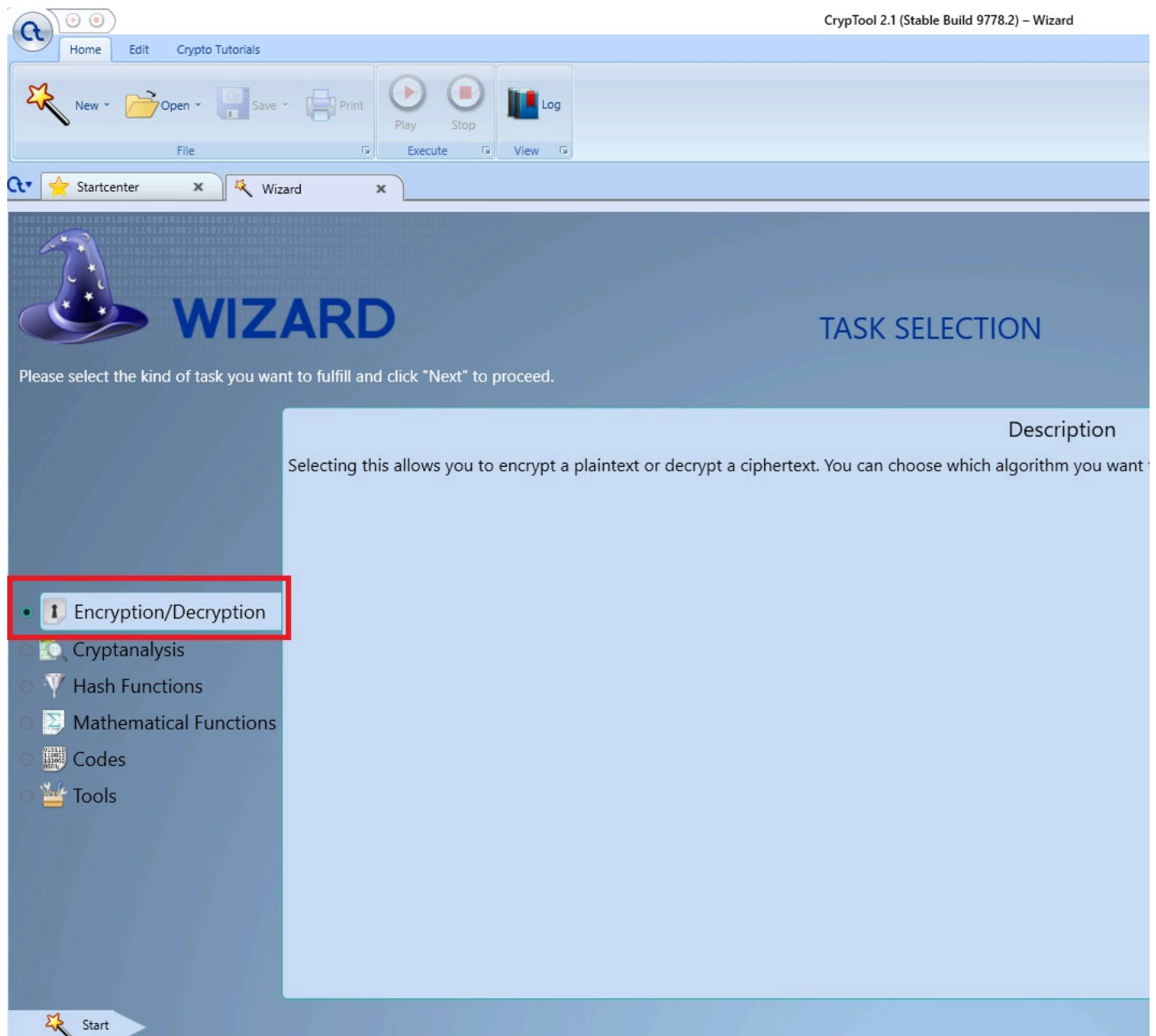
 DES

 DES Cipher

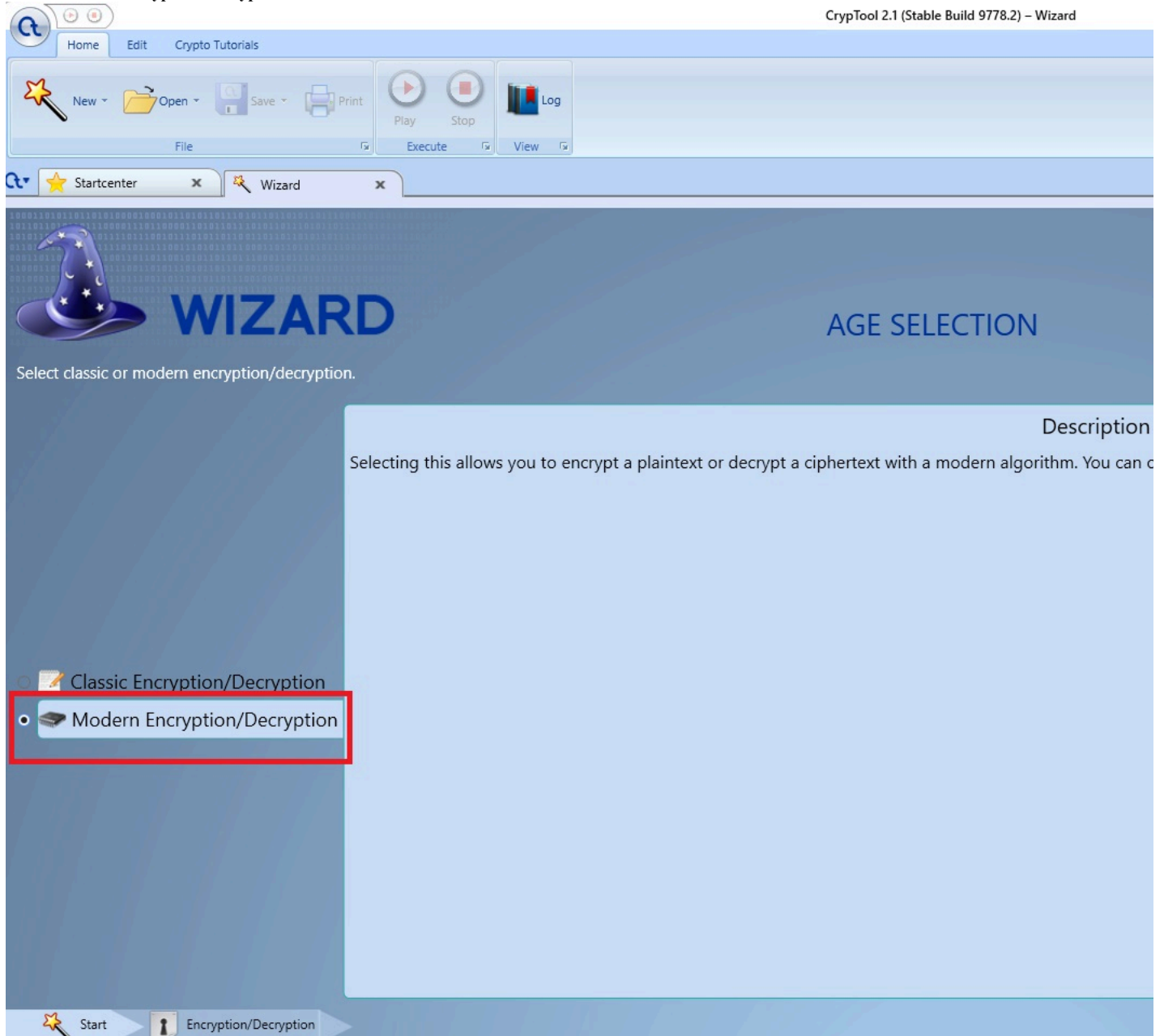
 DES

 DES Cipher

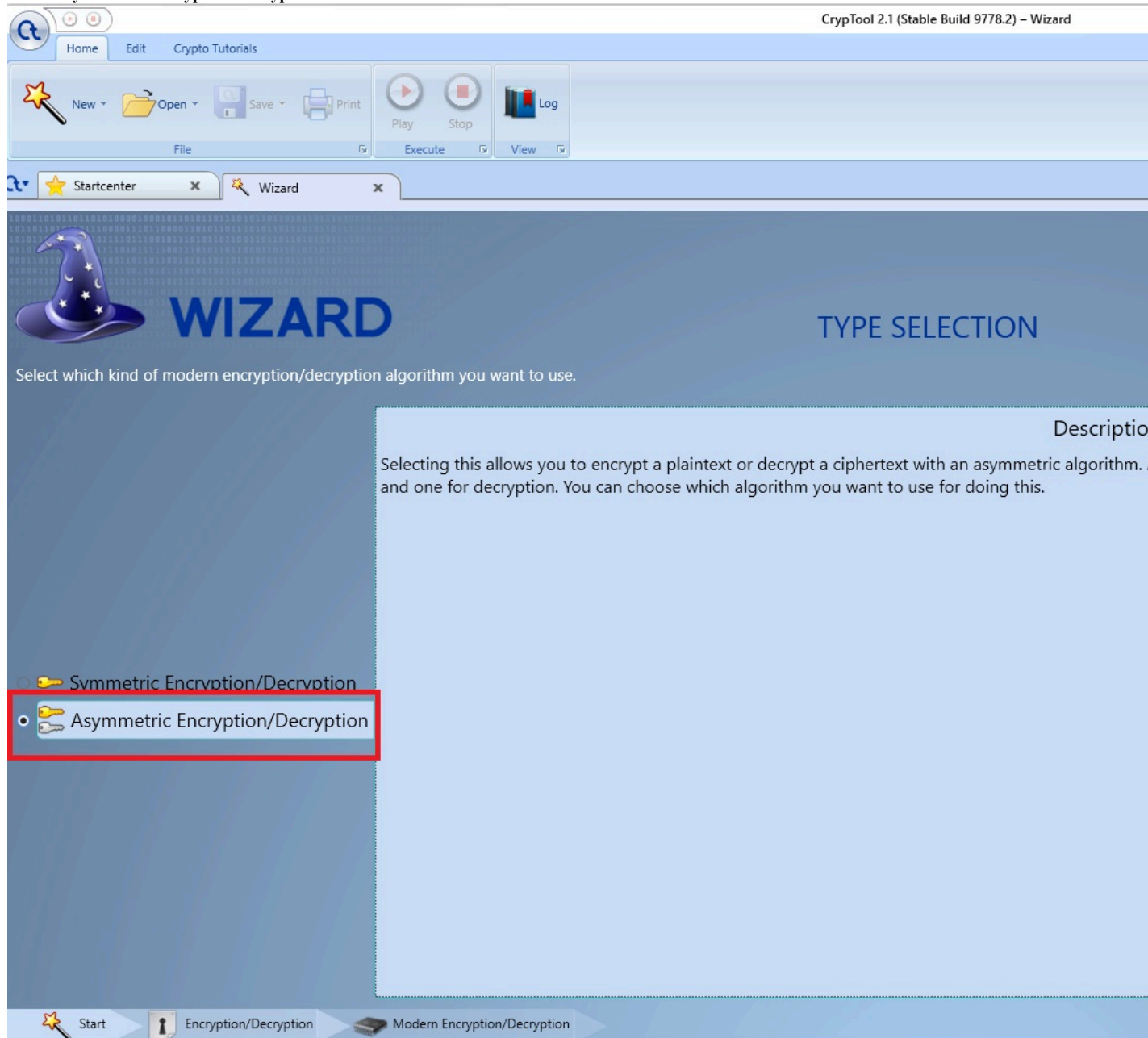
- Select **Encryption/Decryption** from the list of options. Click **Next**.



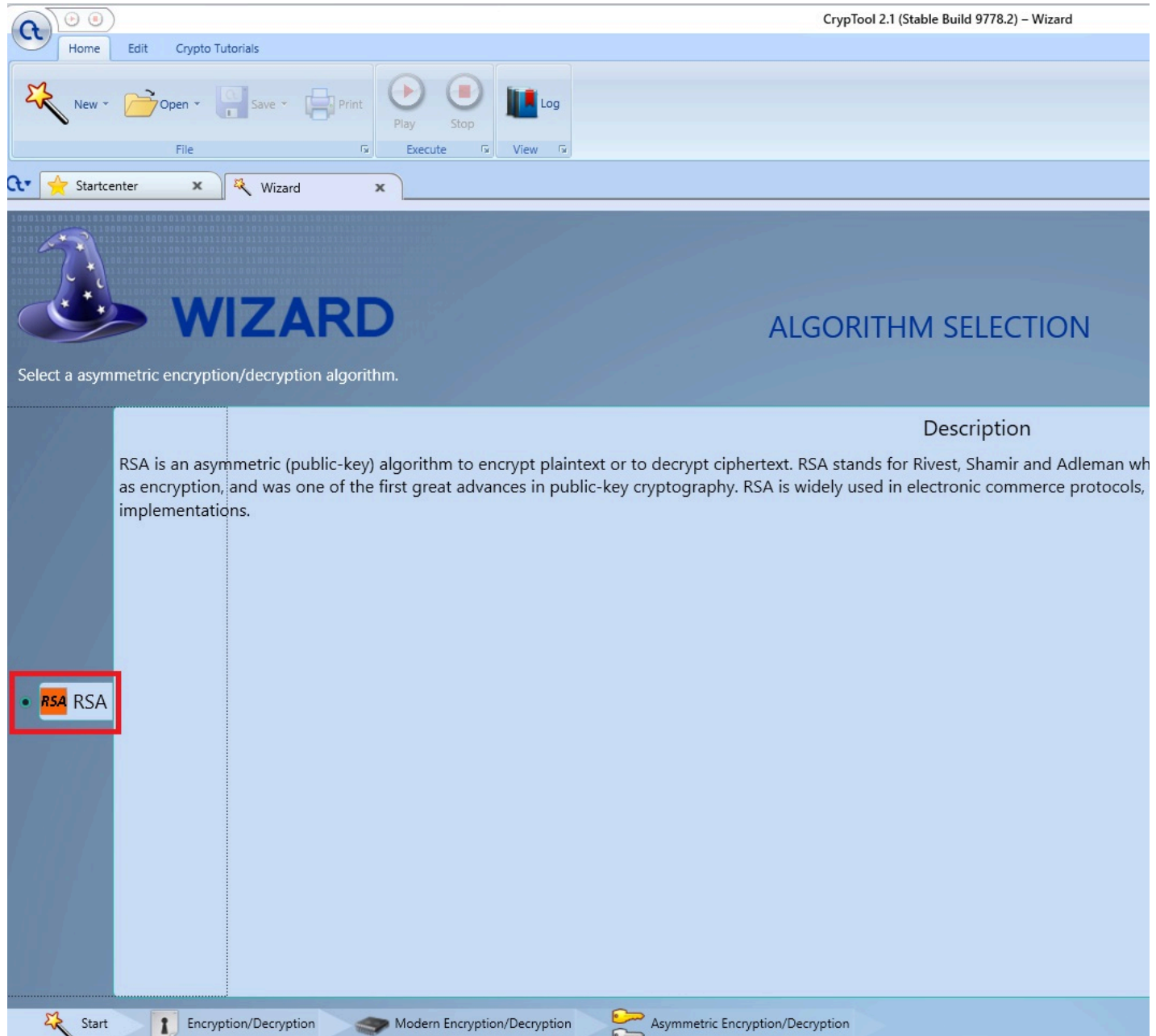
- Select **Modern Encryption/Decryption**. Click Next.



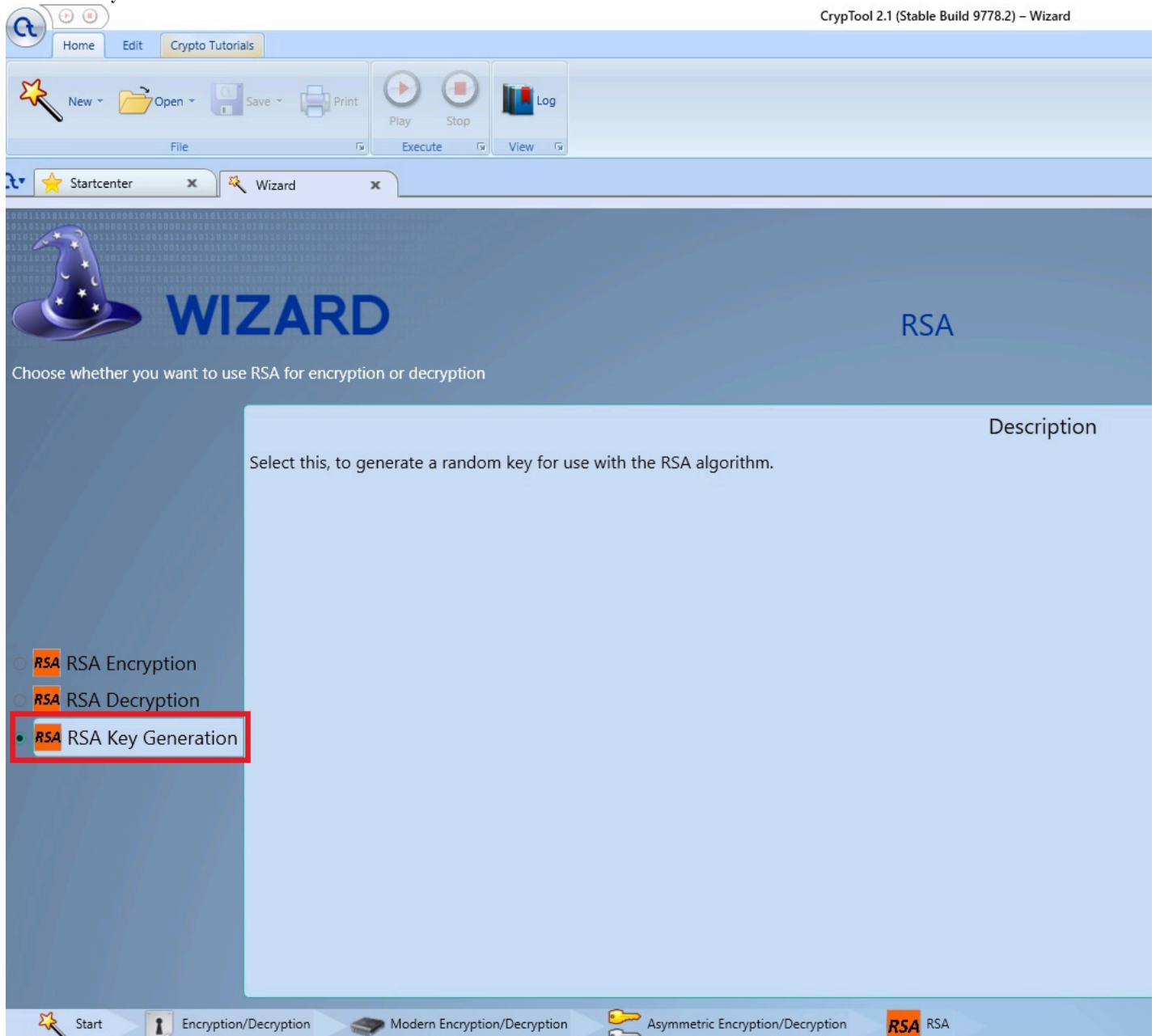
- Select **Asymmetric Encryption/Decryption**. Click **Next**.



- Select **RSA** and click **Next**.



- Select **RSA Key Generation**. Click **Next**.



- The RSA key pair is generated. (N,e) is the public key, and (N,d) is the private key.

Cryptool 2.1 (Stable Build 9778.2) – Wizard

Home Edit Crypto Tutorials

New Open Save Print Play Stop Log

File Execute View

Startcenter Wizard

WIZARD

RSA KEY GENERATION

Here you can see a random generated RSA keypair. (N, e) is the public key and (N, d) is the private key.

N (common part of keypair):

```
385120418848474035014908049463400688173557327275213108164062643494315032848008054004194890431161798153505645714638930288843659922516482010373428555732158257445
266975216938647319929388460136899318246988401647058679805236376664951926044862890426305152922615300535888235043603580070846887132920067999582980936446306227673
188416593471566631597027213778760994131481071271402845049768157390898684080826917603347679188454125280185283
```

616 decimal digits, 2,045 bit

e (public part of keypair):

```
11
```

2 decimal digits, 4 bit

d (private part of keypair):

```
350109471680430940922643681330364261975961206613830098330966039540286393498189140003813536755601634685005132467853572989857872656833165463975844141574689324950
969977469944224836299444054669908471133625819679144253235426365582431567854294424630630619713611417628356895358726192334139932633288682876663122088574913366131
89786853357840546803950371637884911264036684651625424848382126409964023918121472267240353093277034988821451
```

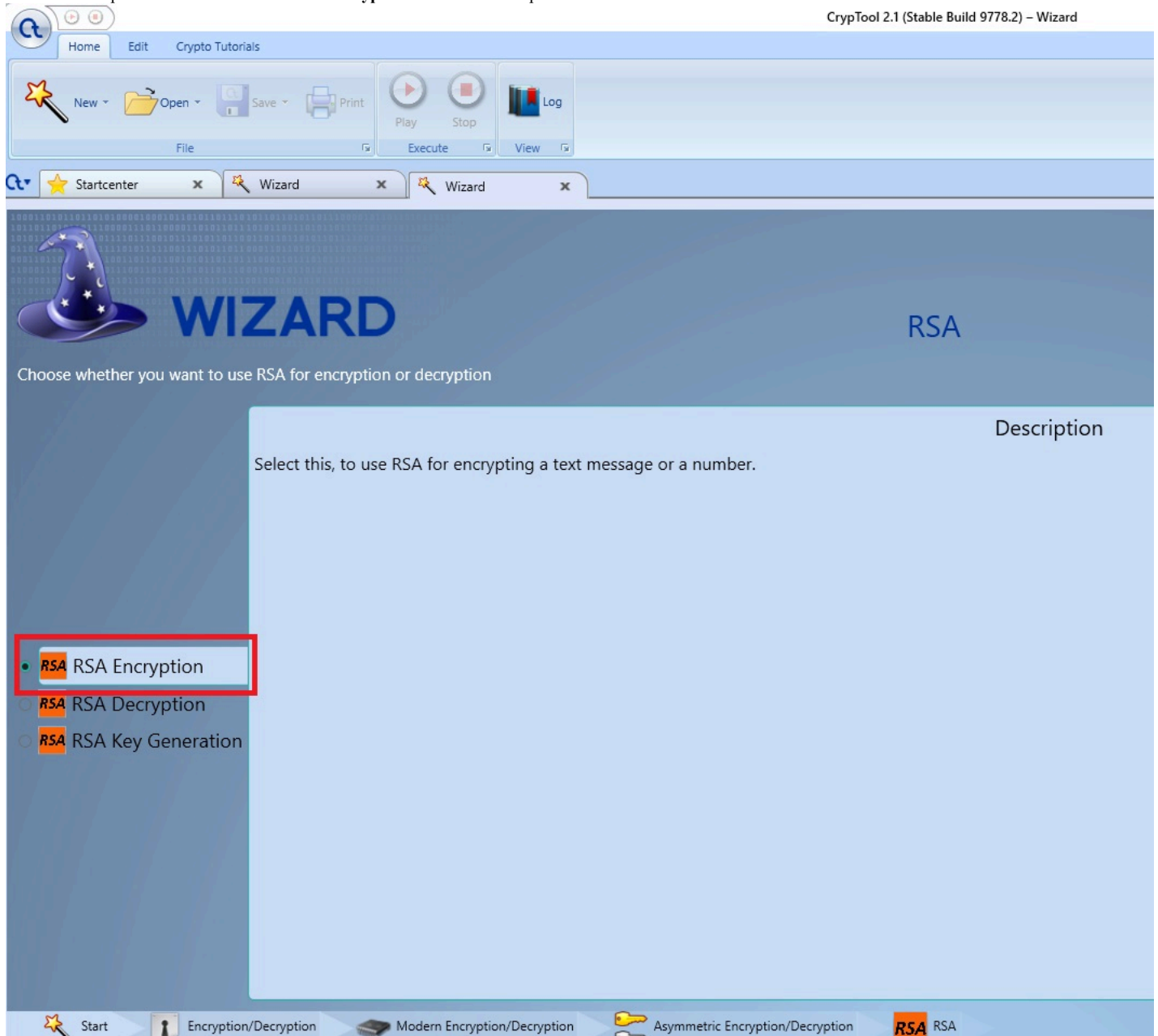
615 decimal digits, 2,042 bit

Start Encryption/Decryption Modern Encryption/Decryption Asymmetric Encryption/Decryption RSA RSA RSA RSA Key Generation

- Select Next.

Step 2: Encryption using public key generated using RSA algorithm

- Click **New** to open a new wizard. Select **RSA Encryption** from the list of options. Click **Next**



- Copy the value of **N** and **e** generated in *Step 1* and paste in the *N* and *e* text box respectively. Type some text in *Text Message to Encrypt*. Click **Next**.

CrypTool 2.1 (Stable Build 9778.2) – Wizard

HomeEditCrypto Tutorials

NewOpenSavePrint

PlayStopLog

FileExecuteView

StartcenterWizardWizard



WIZARD

RSA ENCRYPTION INPUT

Here, you can input the plaintext and the public key to use for encryption.

N (common part of keypair):

38512041884847403501490804946340068817355732727521
31081640626434943150328480080540041948904311617981
53505645714638930288843659922516482010373428555732
15825744573574755465272760495654067898303161637563
71175881799406800115061005926233635880173100366018
69392669752169386473199293884601368993182469884016
47058679805236376664951926044862890426305152922615
30053588823504360358007084688713292006799958298093
64463062276734662212817160484866296562566585140829
45281229893615154689269693266852673448674142149183
81795872188416593471566631597027213778760994131481
07127140284504976815739089868408082691760334767918
8454125280185283

e (public part of keypair):

11

Text message to encrypt:

In cryptography, RSA (which stands for Rivest, Shamir and Adleman who first publicly described it) is an algorithm for public-key cryptography.[1] It is the first algorithm known to be suitable for :
cryptography. RSA is widely used in electronic commerce protocols, and is believed to be secure given sufficiently long keys and the use of up-to-date implementations.

StartEncryption/DecryptionModern Encryption/DecryptionAsymmetric Encryption/DecryptionRSA RSA RSA RSA Encryption

about:blank

10/15

- The Ciphertext (in Hex) is generated.

CrypTool 2.1 (Stable Build 9778.2) – Wizard

Home Edit Crypto Tutorials

New Open Save Print Play Stop Log

File Execute View

Startcenter Wizard Wizard

WIZARD

RSA ENCRYPTION OUTPUT

Here, you can see the RSA encryption output.

Ciphertext (in Hex):

```
4D B4 A8 37 2E DA A9 BD 69 8B 4F 09 A2 8D 1A 90 1E 11 71 81 F0 DC AF 55 58 E6 F0 BC BA 10 C2 86 1C E9 F1 3E BB 85 36 EC 66 62 4E 8A 41 9B 25 6E 3B 4D D9 2C 71 27 22 5C 7B C1 A5 B2 93 B7
1F 93 AF 42 8C 6E C9 12 67 F9 F2 B4 A2 85 41 8F 3D CF 9C 17 45 42 3F BF AC 40 BF 9B 23 99 5F 72 3E C0 6F 51 94 C6 78 BB 2A EF 65 B1 E5 9E 78 34 9E 1A 0A B6 0B 25 BB BC 3C DC C6 C6 16 A4 A
44 E8 69 17 7E 6F 53 03 CC B0 E8 97 B2 03 D2 9C 56 0B 6F 33 E7 40 7D 84 7E 97 9B 8D 02 75 B1 69 04 BD A4 13 FD EA 7C D7 4D A1 D3 B6 E9 7B 0D ED E6 2A 3B 4B 8E CD 4F 52 30 1E D5 DC 95 9
6A 45 9C C5 40 07 E2 5D D4 A1 98 0D 1D 29 1B B6 E5 30 9B 29 77 3B 07 BD E9 02 95 F6 C5 E1 BD DB 9F E5 E0 E9 2F AD FC E4 14 5C BB F6 DF 9C 66 F0 57 06 E9 5A D5 D5 A5 AF D4 5F 90 23 EE E2
7F 2D 18 42 AC A7 11 8F DC 00 AD ED 0E 3E 8D 1E 2A BF D4 54 DE 81 14 E5 8D B1 DA 32 61 4F 11 C3 F9 0B 36 A9 2D 28 D6 F0 E5 B2 2C BD 54 58 C9 2B 92 B2 71 B8 C2 A4 75 A5 87 55 15 DF D4
81 CE 58 FB 63 1C 4F 0E D1 CD A4 2D 66 84 7A D4 26 60 6D 2D C6 53 D8 72 34 95 8D 46 1D DF 73 55 92 D8 9C FB 4D DE AB 98 EE F5 29 BE 8B D0 55 F2 ED 8B 09 BE 70 29 ED E7 16 49 E5 A1 BA 9
```

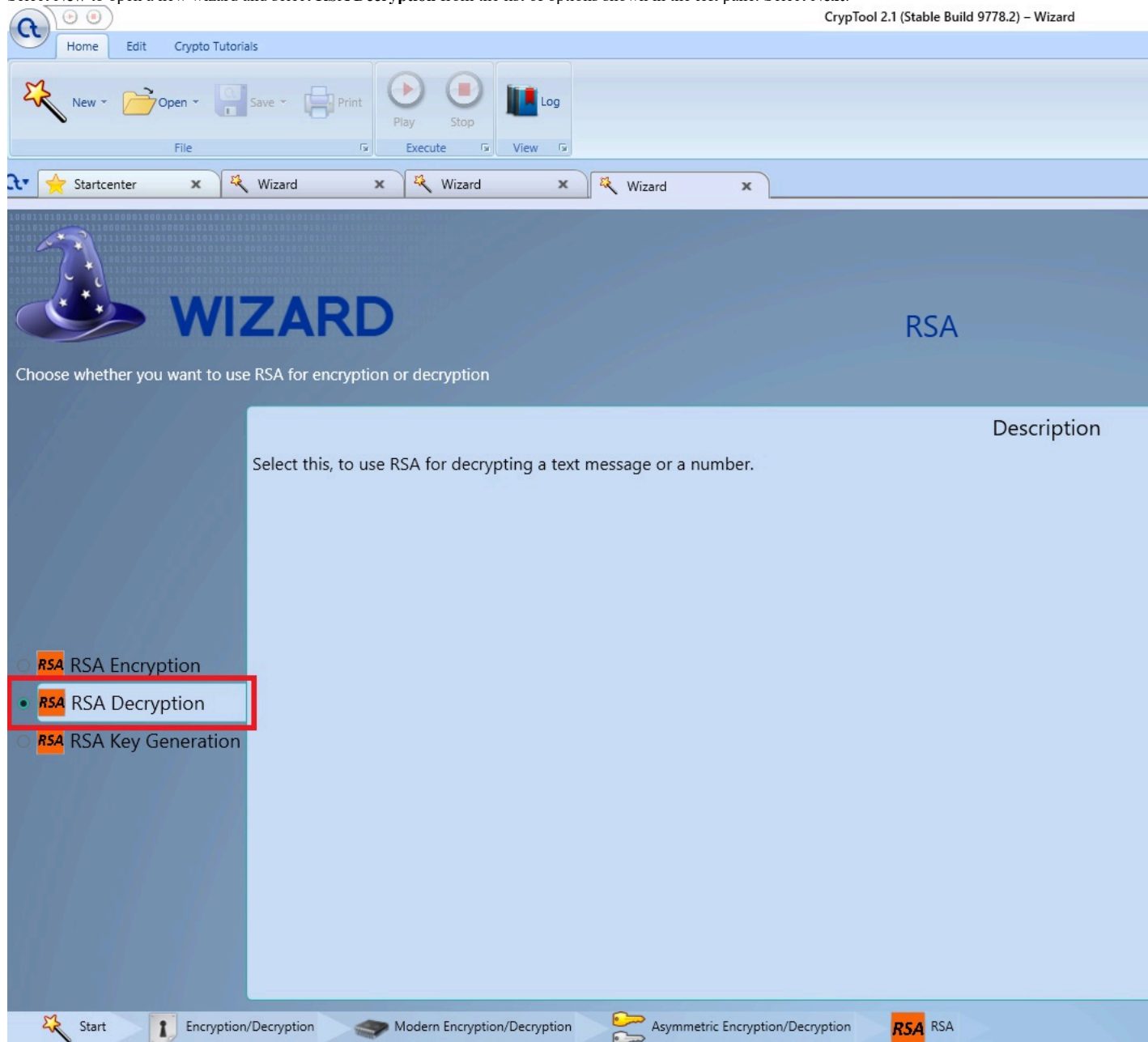
1,535 characters, 1 line

Start Encryption/Decryption Modern Encryption/Decryption Asymmetric Encryption/Decryption RSA RSA RSA RSA Encryption

- Select Next.

Step 3: Decryption using private key generated using RSA algorithm

- Select **New** to open a new wizard and select **RSA Decryption** from the list of options shown in the left pane. Select **Next**.



- Copy the Value of **N** and **d** generated in *Step 1* and paste in the *N* and *d* text box respectively. Similarly, copy the **Ciphertext (in Hex)** generated in *Step 2* and paste in the *Text Message to Decrypt* box. Click on **Next**.

CrypTool 2.1 (Stable Build 9778.2) – Wizard

HomeEditCrypto Tutorials

NewOpenSavePrint

PlayStopLog

FileExecuteView

StartcenterWizardWizardWizardWizard



WIZARD

RSA DECRYPTION INPUT

Here, you can input the plaintext and the private key to use for decryption.

N (common part of keypair):

38512041884847403501490804946340068817355732727521
31081640626434943150328480080540041948904311617981
53505645714638930288843659922516482010373428555732
15825744573574755465272760495654067898303161637563
71175881799406800115061005926233635880173100366018
69392669752169386473199293884601368993182469884016
47058679805236376664951926044862890426305152922615
30053588823504360358007084688713292006799958298093
64463062276734662212817160484866296562566585140829
45281229893615154689269693266852673448674142149183
81795872188416593471566631597027213778760994131481
07127140284504976815739089868408082691760334767918
8454125280185283

d (private part of keypair):

35010947168043094092264368133036426197596120661383
00983309660395402863934981891400038135367556016346
85005132467853572989857872656833165463975844141574
68932495066886141332066145905140061725730146943239
73796256181278909195510005387485123527430091241835
17629699774699442248362994440546699084711336258196
79144253235426365582431567854294424630630619713611
41762835689535872619233413993263328868287666312208
85749133661313944155565639941074681409133579648233
74045050177436133978959203996484068155190778332579
64844021897868533578405468039503716378849112640366
84651625424848382126409964023918121472267240353093
277034988821451

Text message to decrypt (in Hex):

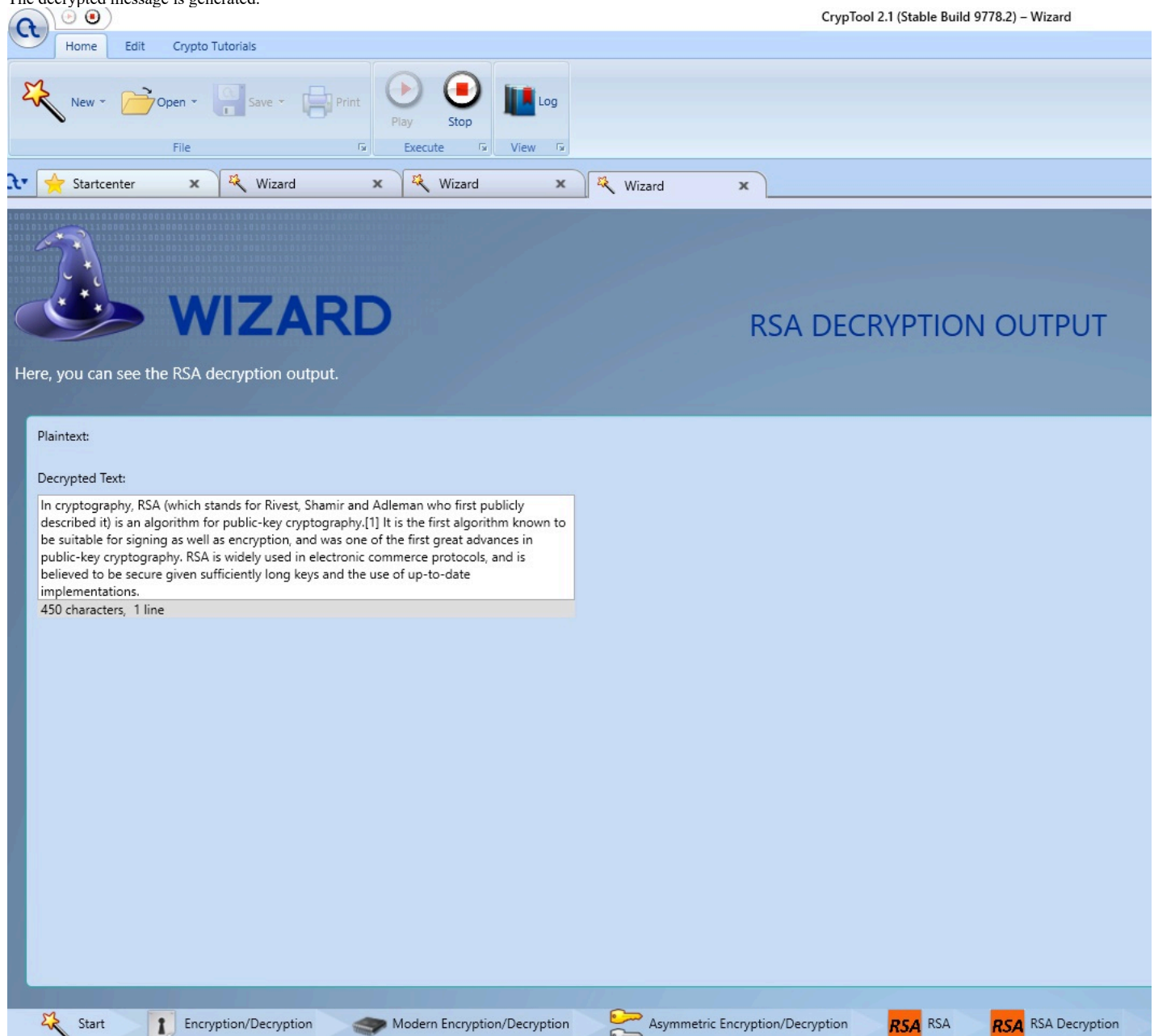
4D B4 A8 37 2E DA A9 BD 69 8B 4F 09 A2 8D 1A 90 1E 11 71 81 F0 DC AF 55 58 E6 F0 BC BA 10 C2 86 1C E9 F1 3E BB 85 36 EC 66 62 4E 8A 41 9B 25 6E 3B 4D D9 2C 71 27 22 5C 7B C1 A5 B2 93 B
1F 93 AF 42 8C 6E C9 12 67 F9 F2 B4 A2 85 41 8F 3D CF 9C 17 45 42 3F BF AC 40 BF 9B 23 99 5F 72 3E C0 6F 51 94 C6 78 BB 2A EF 65 B1 E5 9E 78 34 9E 1A 0A B6 0B 25 BB BC 3C DC C6 C6 16 A4
44 E8 69 17 7E 6F 53 03 CC 80 E8 97 B2 03 D2 9C 56 0B 6F 33 E7 40 7D 84 7E 97 9B 8D 02 75 B1 69 04 8D A4 13 F0 EA 7C D7 4D A1 D3 B6 E9 7B 0D ED E6 2A 3B 48 8E CD 4F 52 30 1E D5 DC 95 9
6A 45 9C C5 40 07 E2 D5 D4 A1 98 0D 1D 29 1B B6 E5 30 9B 29 77 3B 07 BD E9 02 95 F6 C5 E1 BD DB 9F E5 E0 E9 2F AD FC E4 14 5C BB F6 DF 9C 66 F0 57 06 E9 5A D5 D5 A5 AF D4 5F 90 23 EE E
7F 2D 18 42 AC A7 11 8F DC 00 AD ED 0E 3E 8D 1E 2A BF D4 54 DE 81 14 E5 8D B1 DA 32 61 4F 11 C3 F9 0B 36 A9 2D 28 D6 F0 E5 B2 2C BD 54 58 C9 2B 92 B2 71 B8 C2 A4 75 A5 87 55 15 DF D4
81 CE 58 FB 63 1C 4F 0E D1 CD A4 2D 66 84 7A D4 26 60 6D 2D C6 53 D8 72 34 95 8D 46 1D DF 73 55 92 DB 9C FB 4D DE AB 98 EE F5 29 BE 8B D0 55 F2 ED 8B 09 BE 70 29 ED E7 16 49 E5 A1 BA

StartEncryption/DecryptionModern Encryption/DecryptionAsymmetric Encryption/DecryptionRSA RSARSA RSA Decryption

about:blank

13/15

- The decrypted message is generated.



Exercise

- Here are two sets of (N), (e), and (d) values for an RSA experiment. These values represent the modulus ((N)), the public exponent ((e)), and the private exponent ((d)). (N,e) is the public key and (N,d) is private key. Encrypt and decrypt the message using these key pairs.

Set 1:

- (N = 3233)
- (e = 17)
- (d = 2753)

Set 2:

- (N = 2773)
- (e = 13)
- (d = 937)

Summary

In this reading, you have learned to successfully install **CrypTool 2.1 (Stable Build 9778.2)**, generate RSA keys, and perform encryption and decryption tasks. This reading provides a solid foundation for more advanced cryptographic experiments.

Author(s)

Dr. Manish Kumar



Skills Network